

Masking cannot remove what is inside the region of interest: location-dependent shortcut mitigation across dermoscopy, ultrasound and capsule endoscopy

Abstract

Medical image classifiers exploit acquisition artifacts—hair, sonographer calipers, intestinal debris—that correlate with the diagnosis, and the standard defence is to restrict the model to the region of interest (ROI). We show that this defence depends on where the artifact sits. In pre-registered traps built from real artifacts in four diseases and three modalities, masking removes an out-of-ROI shortcut but leaves an in-ROI one largely intact (crossover +0.15 to +0.37 reversed-test AUROC, all Holm-significant). Because images with in-ROI and out-of-ROI artifacts differ (for example in lesion size), we confirmed the law by pasting the *same* real artifact inside or outside the ROI of the *same* images: masking then removes the shortcut completely outside the ROI and keeps it, with increased reliance, inside (location interaction hair +0.25, thyroid +0.28, ovary +0.07, capsule +0.50); covariate-matched traps agree. The law holds across encoder families, model sizes, fine-tuning and the zero-shot scores of vision–language models. A linear-Gaussian account explains when in-ROI masking helps a little and when it harms: fitted only to each model’s clean and correlated performance, it matches held-out reversed-test AUROC with a mean absolute error of 0.039, against 0.108 for a symmetry heuristic. On the unaltered, patient-disjoint thyroid test split, masking lowers AUROC on shortcut-conflicting cases and lowers sensitivity at all five pre-registered operating points, most for malignant nodules that carry a caliper. Which remedy works depends on what carries the shortcut—the artifact’s pixels, an appearance that resembles the pathology, or correlates of the artifact—but none transfers uniformly to unaltered data. Audits of ROI-masked models should report artifact location and performance on shortcut-conflicting cases. Code, pre-registrations and results are public.

Keywords: shortcut learning, spurious correlation, artifacts, dermoscopy, chest radiography, concept erasure

1. Introduction

Deep classifiers for medical images frequently rely on features that are correlated with the label in the training distribution but are not evidence of disease [9, 7]. In dermoscopy, surgical skin markings [29] and ruler scale bars [30] changed the melanoma predictions of a market-approved system; in chest radiography, pneumothorax classifiers detect the chest drain that treats the condition rather than the condition itself

[20, 14, 32]. Such correlations are causal in origin—suspicious lesions are measured and marked, collapsed lungs are drained—so they are present in every retrospectively collected dataset.

The most widely recommended defence is to restrict the classifier to the region of interest (ROI): segment the lesion or the lungs and discard the rest of the image. The rationale assumes that the shortcut lives in the background. Rulers, hair, ink and drains do not respect that assumption. When the artifact overlaps the ROI, masking cannot remove it; it removes everything *else*. We ask whether the effect of a mitigation depends on where the artifact sits, which quantity governs that dependence, what supervision each family of mitigations actually needs, and whether the in-ROI regime can be mitigated without localising the artifact.

Contributions.

1. **A location law for ROI masking.** In pre-registered traps built from real artifacts in four diseases and three imaging modalities, masking removes an out-of-ROI shortcut but leaves an in-ROI one largely intact (Holm-significant crossovers in 4/4 cohorts). Because Trap A and Trap B images differ systematically, we establish the law causally with a *real-artifact transplant*—the same real artifact pasted inside or outside the ROI of the same images—and observationally with covariate-matched traps; it replicates across three encoder families, three model sizes, end-to-end fine-tuning in every cohort and the zero-shot scores of vision–language models (Sec. 5).
2. **Mechanism and a closed-form account.** Masking retains an in-ROI artifact while removing competing evidence, so reliance on it grows; a linear-Gaussian model states when in-ROI masking harms and, fitted to each model’s clean and correlated performance only, matches its held-out reversed-test AUROC far better than simple reference predictors (pre-registered; Secs. 6, 7).
3. **Clinical consequences on unaltered data.** On the patient-disjoint official thyroid test split, masking lowers AUROC on shortcut-conflicting cases and lowers sensitivity at every pre-registered operating point, most for malignant nodules that carry a caliper (Sec. 8).
4. **What carries the shortcut decides the fix.** A decision guide that separates distinct overlays, pathology-like artifacts and correlate-carried shortcuts, benchmarked against DFR, JTT, LEACE, SPLINCE, LaMa inpainting and an annotation-free mask-then-erase tool, with the caveat that no remedy transfers uniformly to unaltered data (Sec. 9; methods in the supplement).
5. **Reproducibility.** Pre-registration of every experiment with a registry of commit times, Holm-corrected primary claims, replication of the original pilot (73/88 numbers), a full rebuild on new hardware that reproduces every primary crossover within 0.01, public code, and a CLAIM 2024 checklist.

2. Related work

Artifacts in dermoscopy. Bissoto et al. [3] annotated seven artifact types in ISIC images and showed that

debiasing is harder than it looks; Nauta et al. [19] neutralised coloured calibration patches by inpainting and retraining—an artifact that a colour rule localises and that lies largely outside the lesion. Hair occlusion degrades classifiers trained on HAM10000 [27], and removing dermoscopic dark-corner artifacts is not clearly beneficial, whereas adding synthetic ones to the training set improves classification [22]; both treat an artifact’s presence, not its position relative to the lesion. MaskMedPaint [15] inpaints regions *outside* the classification-relevant area. None of these treats artifact location relative to the ROI as a variable.

Ultrasound. In fetal ultrasound, calipers and on-screen text act as shortcuts even for segmentation [18]; we study them for classification in thyroid and ovarian ultrasound, where they lie on the nodule or tumour.

Chest radiography. Hospital-specific signals limit the generalisation of pneumonia classifiers [32]. Drain-driven pneumothorax shortcuts are documented on NIH-CXR14 and CheXpert [20, 14]; recent work localises them to late layers of probed encoders [21] and mitigates them by prevalence-equalised recalibration [16], which requires the artifact label at test time.

Group robustness and erasure. Reweighting and GroupDRO [24], last-layer retraining [17] and linear concept erasure [23, 1] require artifact labels or paired data; erasure from unpaired labels removes label-correlated signal when the concept is correlated with the label, which is why we build pairs by insertion. Task-preserving linear erasure (SPLINCE; 13) and projection of biased directions in vision–language models [5] are the closest erasure methods; both require concept labels or prompts, whereas our concept subspace is estimated from synthetic insertions. Random erasing and occlusion augmentation [34, 8] use similar overlays as training noise; we show that this does not remove the shortcut. Test-time selection [2] steers dermoscopy models with user keypoints.

Foundation models. Classifiers trained on frozen foundation-model features inherit shortcut-like behaviour that pooled multi-institutional evaluation can hide [10], and shortcut testing [4] probes whether a model’s performance gaps stem from encoding a sensitive attribute. We ask where in the image the shortcut sits and remove it from the frozen representation.

3. Data

Controlled dermoscopy cohort. ISIC 2018 Task 1/2 images with manual lesion masks and melanoma labels from the union of the ISIC 2016 and 2017 tables (three images with conflicting labels dropped); perceptual-hash (Hamming ≤ 8) and lesion grouping; 2,437 images for which a 65×19 px ruler (0.46% of a 518×518 image) can be placed at 0/25/50/75/100% overlap with the lesion within 10 percentage points (train/val/test 1,849/371/371 before common support).

Real hair cohort (ISIC 2019). All 25,331 ISIC 2019 training images; hair/ruler masks from the archive of Wegley et al. [28]; lesion masks manual for HAM10000 [25] and predicted elsewhere by a U-Net trained on ISIC 2018 Task 1. Two traps share one hair-free group (≤ 30 hair pixels at native resolution): Trap A contains hair with overlap $r = |\text{hair} \cap \text{lesion}|/|\text{hair}| \geq 0.5$, Trap B $r < 0.1$. Cells are subsampled to the hair-free source mix (HAM/BCN/MSK) within each label, giving Trap A 704/157/2,738/989 and Trap B 704/157/4,327/930 images (hair-free benign/melanoma, hair benign/melanoma); five seeds of five-fold group-safe cross-validation (leakage groups: lesion identifiers and pHash); 172 reversed-test melanomas per trap and seed.

Thyroid ultrasound. TN3K images [11] with TNCD benign/malignant labels from cytological biopsy [12] (3,493 images of 2,421 patients; 1,210 malignant) and manual nodule masks; the official train/test split (2,879/614 images) is patient-disjoint [12]. Sonographer calipers were segmented by a frozen rule-based detector (dotted measurement lines and ‘+’ ends; audit in Sec. 11). Trap A: calipers with $r \geq 0.5$ (1,150 images); Trap B: $r < 0.1$ (326); caliper-free: 1,801. Patient identifiers are not published, so within the pooled images leakage groups are pHash chains (stricter embedding-based groups in Sec. 10.2).

Ovarian tumour ultrasound. MMOTU OTU_2d [33] (CC BY 4.0): 1,202 images with tumour masks, labelled benign cystic (chocolate cyst, serous cystadenoma, simple cyst) versus lesions with solid components (teratoma, theca-cell tumour, mucinous cystadenoma, high-grade serous carcinoma). The thyroid detector is applied unchanged: Trap A 341/276, Trap B 91/103, caliper-free 170/178 (benign-cystic/solid).

Capsule endoscopy. SEE-AI frames [31] (CC BY 4.0) containing only erosions (3,550) or only polyp-like lesions (1,931); the ROI is the union of the expert lesion boxes. Contamination (debris, bubbles, bile) is segmented by a linear probe on DINOv2 patch tokens trained on 2,160 expert-masked frames (held-out pixel accuracy 0.92, IoU 0.62; expert masks are used where available). Debris-free: $< 3\%$ of the frame; Trap A: debris $\geq 10\%$ with $r \geq 0.5$; Trap B: $r < 0.1$. Leakage groups: blocks of 100 consecutive frame numbers plus pHash duplicates.

Chest radiography. NIH ChestX-ray14 [26]; lung ROIs from the TorchXRyVision PSPNet [6]. *Controlled cohort:* 1,906 pneumothorax images without a chest drain (NEATX annotations) and 5,718 *No Finding* images from other patients; a 100×12 px radio-opaque tube is placed at controlled overlap with the lungs (7,610 images in common support). *Real drains* (in-ROI only): 29,687 images; drains are labelled by NEATX for pneumothorax-positive images and by a drain detector (precision 0.95) for negatives; environments are built within view position (AP/PA); five patient-grouped folds.

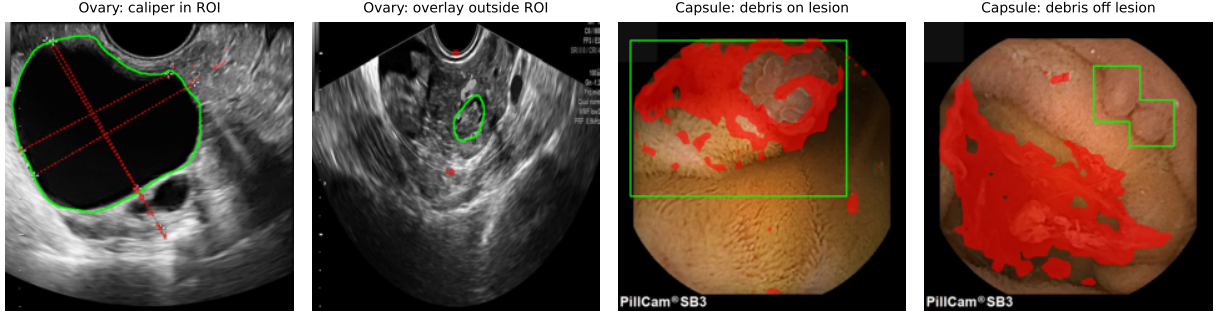


Figure 1: Real in-ROI and out-of-ROI artifacts (openly licensed cohorts). Green: ROI; red: artifact mask. Left: MMOTU ovarian tumours (caliper inside the tumour; overlay outside). Right: SEE-AI capsule frames (debris on and off the lesion box).

4. Problem setup and protocol

Let $Y \in \{0, 1\}$ be the diagnosis and $A \in \{0, 1\}$ the presence of an artifact. A *trap* trains with $P(A=1 | Y=1) = 0.9$ and $P(A=1 | Y=0) = 0.1$ and tests on a correlated environment with the same rates, a *reversed* environment with the rates swapped (0.1/0.9), and a clean environment in which the artifact is independent of the label (real traps: present in half of each class; controlled sweeps: absent). A model that uses A gains on the correlated test and collapses on the reversed one; the primary metric is reversed-test AUROC, and clean AUROC measures the cost of a mitigation. The overlap $r = |\text{artifact} \cap \text{ROI}| / |\text{artifact}|$ is the controlled variable. Three designs vary it: *controlled sweeps* paste a synthetic artifact at overlaps 0–100% on artifact-free images; *real traps* compare images whose real artifact lies inside the ROI ($r \geq 0.5$, Trap A) or outside it ($r < 0.1$, Trap B), each added to one shared artifact-free group; the *real-artifact transplant* pastes one real artifact instance, cut out with its mask, inside ($r \geq 0.95$) or outside ($r = 0$) the ROI of the same artifact-free images. The *location crossover* $[\text{mask} - \text{ERM}]_{\text{out}} - [\text{mask} - \text{ERM}]_{\text{in}}$ of reversed-test AUROC summarises each design. Unless stated otherwise backbones are frozen and linear heads are fitted (end-to-end fine-tuning in Sec. 5.5); regularisation strength and the decision threshold (balanced accuracy) are selected on clean validation data only. Uncertainty is a hierarchical paired bootstrap that resamples training seeds (or cross-validation folds), then paired test images within each, computes the AUROC difference inside each cluster and averages; predictions of different models are never pooled into one ROC curve. Splits are grouped by lesion/patient identifier where published and otherwise by perceptual hash (and frame blocks); a stricter embedding-based grouping is a sensitivity analysis (Sec. 10.2).

Table 1: Synthetic ruler, ISIC 2018, DINOv2 ViT-B/14 @518: AUROC and reversed-test Δ vs ERM [95% CI]

Method	Needs	clean@0	rev@0	clean@50	rev@50	clean@100	rev@100	Δ rev@0	Δ rev@50	Δ rev@100
ERM	—	0.797	0.665	0.793	0.573	0.792	0.515			
ROI masking	ROI mask (train+test)	0.849	0.849	0.829	0.476	0.825	0.386	+0.184 [+0.146, +0.224]	-0.097 [-0.138, -0.056]	-0.129 [-0.170, -0.087]
Inpainting (oracle)	artifact pixels (train+test)	0.810	0.806	0.809	0.783	0.807	0.759	+0.141 [+0.120, +0.162]	+0.211 [+0.187, +0.234]	+0.243 [+0.218, +0.269]
Group-balanced	image-level A (train)	0.814	0.802	0.815	0.792	0.806	0.796	+0.137 [+0.109, +0.168]	+0.219 [+0.188, +0.252]	+0.281 [+0.248, +0.315]
DFR	image-level A (val)	0.786	0.789	0.787	0.794	0.789	0.794	+0.124 [+0.091, +0.160]	+0.221 [+0.184, +0.261]	+0.279 [+0.240, +0.320]
LEACE (paired, oracle renders)	paired renders (train)	0.829	0.832	0.810	0.810	0.809	0.809	+0.167 [+0.139, +0.197]	+0.237 [+0.212, +0.263]	+0.294 [+0.265, +0.325]
Repair consistency	artifact pixels (train)	0.828	0.707	0.822	0.616	0.814	0.558	+0.042 [+0.015, +0.069]	+0.043 [+0.014, +0.072]	+0.042 [+0.012, +0.073]
Insert-then-erase (artifact templates)	artifact templates	0.848	0.844	0.850	0.833	0.847	0.818	+0.179 [+0.152, +0.206]	+0.260 [+0.229, +0.291]	+0.303 [+0.271, +0.336]
Insert-then-erase + balanced (artifact templates)	artifact templates + A (train)	0.841	0.842	0.843	0.840	0.844	0.842	+0.177 [+0.151, +0.205]	+0.267 [+0.238, +0.298]	+0.327 [+0.293, +0.362]
Insert-then-erase rank-1 (ablation)	artifact templates	0.811	0.799	0.809	0.770	0.806	0.730	+0.134 [+0.115, +0.154]	+0.197 [+0.174, +0.220]	+0.215 [+0.190, +0.240]
Insertion augmentation	artifact templates	0.814	0.726	0.812	0.654	0.810	0.603	+0.061 [+0.049, +0.074]	+0.081 [+0.068, +0.094]	+0.088 [+0.077, +0.099]

5. Results: a location law for ROI masking

5.1. Controlled dermoscopy: the effect reverses with overlap

With a synthetic ruler whose overlap with the lesion is swept from 0 to 100% (2,437 ISIC 2018 images, DINOv2 ViT-B/14 at 518 px, three seeds), ROI masking improves reversed-environment AUROC over ERM by +0.184 [+0.146, +0.224] when the ruler lies outside the lesion and *lowers* it by $-0.097 [-0.138, -0.056]$, $-0.112 [-0.155, -0.068]$ and $-0.129 [-0.170, -0.087]$ at 50, 75 and 100% overlap (location interaction +0.313 [+0.272, +0.353]; Table 1). The sign flip is reproduced at 224 px and under randomised ruler colour, opacity, ticks and scale. At 0% overlap the masked model’s correlated and reversed AUROC coincide (0.849/0.849); at 100% its correlated AUROC exceeds ERM’s (0.961 vs 0.937): it uses the ruler *more*. Synthetic calipers on thyroid nodules, synthetic debris on capsule lesions and synthetic calipers on ovarian tumours give the same pattern (location interactions +0.432 [+0.377, +0.492], +0.205 [+0.150, +0.259]; ovary at full overlap $-0.197 [-0.292, -0.100]$; Fig. ??).

5.2. Real artifacts in four diseases (pre-registered)

Each real trap pair shares one artifact-free group; Trap A adds images whose artifact lies mostly inside the ROI ($r \geq 0.5$), Trap B images whose artifact lies outside it ($r < 0.1$); five seeds of five-fold group-safe cross-validation. The crossover $[\text{mask} - \text{ERM}]_{\text{Trap B}} - [\text{mask} - \text{ERM}]_{\text{Trap A}}$ is positive and Holm-significant in all four cohorts: dermoscopic hair +0.145 [+0.105, +0.181], thyroid calipers +0.230 [+0.204, +0.255], ovarian calipers +0.170 [+0.116, +0.224] and capsule debris +0.368 [+0.340, +0.397]. On dermoscopy masking *harms* in-ROI ($-0.037 [-0.057, -0.019]$, negative in all five seeds; BCN $-0.051 [-0.075, -0.027]$) and helps out of the ROI (+0.107 [+0.078, +0.135]). On thyroid, ovarian and capsule images in-ROI masking still helps, but far less than out of the ROI (thyroid +0.111 [+0.097, +0.124] versus +0.341 [+0.320, +0.363]; capsule +0.096 versus +0.465), and the masked Trap A model still relies on the calipers (correlated 0.902 versus reversed 0.400). What is universal in our data is therefore the *gap*: masking leaves an in-ROI shortcut largely intact while it removes an out-of-ROI one. Harm is conditional; it appears when the mask also removes disease context (Sec. 7).

Table 2: The location law under three designs (reversed-test AUROC, DINOv2 ViT-B/14; 95% hierarchical bootstrap CIs). *Real traps*: crossover $[\text{mask} - \text{ERM}]_{\text{out}} - [\text{mask} - \text{ERM}]_{\text{in}}$ between different artifact-bearing images (rebuilt run). *Matched*: the same after 1:1 propensity matching of the in-ROI and out-of-ROI artifact-bearing images on lesion size, artifact amount and acquisition covariates (max $|\text{SMD}|$ after matching). *Transplant*: the same real artifact instance pasted inside or outside the ROI of the same artifact-free images (location interaction). Holm over four cohorts per design.

Cohort	Real traps	Matched	$ \text{SMD} _{\text{max}}$ before/after	Transplant	n transplants
ISIC hair	+0.147 [+0.119, +0.177]	+0.154 [+0.125, +0.184]	2.13 / 0.11	+0.253 [+0.231, +0.275]	860
Thyroid	+0.239 [+0.212, +0.264]	+0.238 [+0.207, +0.269]	0.53 / 0.04	+0.277 [+0.264, +0.290]	1,800
Ovary	+0.170 [+0.116, +0.223]	+0.163 [+0.087, +0.237]	0.39 / 0.08	+0.074 [+0.055, +0.092]	348
Capsule	+0.377 [+0.339, +0.412]	+0.230 [+0.125, +0.337] [†]	2.57 / 0.23	+0.497 [+0.455, +0.539]	340

[†]Fewer than 30 matched pairs per label (pre-registered feasibility limit): exploratory only. Capsule lesions carrying debris are much larger than those with debris beside them.

5.3. Same images, same artifact: real-artifact transplant

Trap A and Trap B contain different artifact-bearing images, and these differ systematically (Table ??): lesions carrying in-lesion hair are much larger (standardised mean difference 2.1), as are thyroid nodules with calipers on them (0.4) and capsule lesions with debris on them (2.6), and in-ROI calipers carry more marker pixels. The crossover could therefore reflect population differences rather than artifact location. We removed this possibility by design (pre-registered): real artifact instances—a caliper measurement line with its end crosses, a hair strand, a piece of debris—were cut from artifact-bearing images with their masks and pasted *inside* (overlap ≥ 0.95) or *outside* (overlap 0) the ROI of the *same* artifact-free images, with the same instance, transform and compositing in both locations; environments as in the controlled sweeps. With the artifact outside the ROI masking removes the shortcut entirely—the masked model’s reversed, correlated and clean AUROC coincide—whereas with the identical artifact inside the ROI the shortcut survives masking (Table ??). The location interaction is positive in 4 of 4 cohorts after Holm correction: dermoscopic hair +0.253 [+0.231, +0.275], thyroid +0.277 [+0.264, +0.290], ovary +0.074 [+0.055, +0.092], capsule +0.497 [+0.455, +0.539] (Table 2). Inside the ROI the masked model reacts to inserting the artifact at least as strongly as ERM does (same-head $|\Delta p|$, mask versus ERM: dermoscopic hair 0.297 versus 0.340; thyroid 0.271 versus 0.206; ovary 0.116 versus 0.037; capsule 0.551 versus 0.238), whereas group balancing is nearly location-invariant, as the linear-Gaussian account predicts (interaction dermoscopic hair $-0.052 [-0.066, -0.039]$, thyroid $-0.016 [-0.021, -0.010]$, ovary $-0.011 [-0.020, -0.003]$, capsule $-0.032 [-0.054, -0.009]$).

5.4. Covariate-matched traps

As a second, observational check we matched the artifact-bearing images of the two traps 1:1 on a propensity score of lesion or nodule size, artifact amount and acquisition covariates (age, sex, body site, source,

diagnosis subtype for dermoscopy; image size and grey level for ultrasound; pre-registered), leaving everything else unchanged. The matched crossovers are dermoscopic hair $+0.154$ [$+0.125, +0.184$] (unmatched $+0.147$; max |SMD| $2.13 \rightarrow 0.11$); thyroid $+0.238$ [$+0.207, +0.269$] (unmatched $+0.239$; max |SMD| $0.53 \rightarrow 0.04$); ovary $+0.163$ [$+0.087, +0.237$] (unmatched $+0.170$; max |SMD| $0.39 \rightarrow 0.08$), all Holm-significant. For capsule matching left only 19 pairs in one label, below the pre-registered minimum of 30, so its matched estimate ($+0.230$ [$+0.125, +0.337$]) is exploratory; there the transplant is the decisive test.

5.5. Replication across encoders, training regimes and foundation models

The crossover replicates with MedSigLIP (thyroid $+0.163$ [$+0.132, +0.194$], capsule $+0.332$ [$+0.296, +0.365$], ovary $+0.179$), a CNN (ConvNeXt-Base: thyroid $+0.284$ [$+0.253, +0.315$], capsule $+0.292$ [$+0.248, +0.337$]) and DINOv2 at three sizes (positive in all nine cohort-size combinations; Supplement S7), and with end-to-end fine-tuning (crossover with end-to-end fine-tuning: capsule (ResNet-50, 5 clusters) $+0.584$ [$+0.469, +0.691$]; ovary (ResNet-50, 5 clusters) $+0.136$ [$-0.085, +0.329$]; thyroid (ResNet-50, 5 clusters) $+0.057$ [$-0.021, +0.130$]; thyroid (ViT-S, 5 clusters) $+0.168$ [$+0.074, +0.265$]). Without any task training, the zero-shot scores of vision-language models are already swayed by artifacts, and masking changes them as the law predicts (MedSigLIP crossovers: thyroid $+0.103$ [$+0.079, +0.127$], capsule $+0.049$ [$+0.024, +0.076$], ovary $+0.181$ [$+0.135, +0.228$]; DermLIP on hair $+0.051$ [$+0.031, +0.071$]); MedSigLIP’s zero-shot thyroid diagnosis itself is at chance (clean AUROC 0.45), so we use these scores only to show that the effect is a property of the representation.

The one encoder without a hair gap. With the dermatology foundation model DermLIP and a trained linear head, masking harms at *both* hair locations (Trap A -0.085 [$-0.107, -0.063$], Trap B -0.087 [$-0.115, -0.060$]) and the hair crossover vanishes (-0.002 [$-0.033, +0.028$]), although the controlled ruler sweep reverses with DermLIP as with DINOv2 ($+0.161$ at no overlap, -0.263 to -0.309 at 25–100%). Two measurements explain it. First, with DermLIP masking costs disease signal wherever the hair lies (clean AUROC -0.060 [$-0.074, -0.045$] in Trap A and -0.081 [$-0.098, -0.064$] in Trap B): this encoder, trained on whole dermatology images, loses more from a masked background than DINOv2. Second, the pixel part of its hair shortcut is small in both traps (oracle hair inpainting $+0.044$ [$+0.034, +0.053$] and $+0.045$ [$+0.026, +0.059$]), whereas balancing, which also removes correlates, gains $+0.143$ and $+0.119$: masking removes little of the shortcut at either location, so there is no gap to find. The linear-Gaussian account fitted to DermLIP’s clean and correlated AUROC anticipates this (-0.003 ; Sec. 7). The law describes what masking can remove; when the shortcut is carried by correlates and the encoder is fragile to masking, masking removes little at either location.

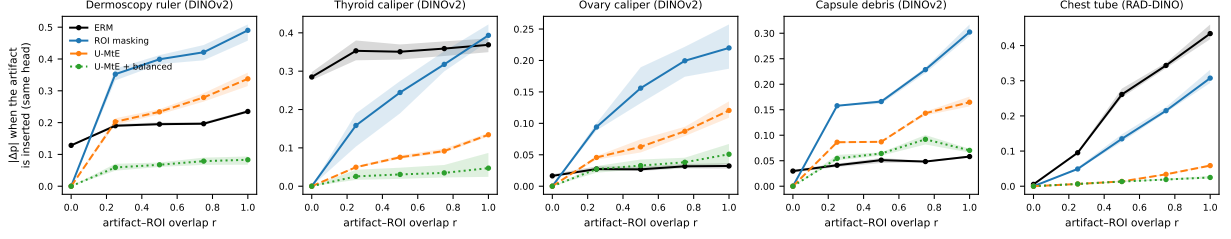


Figure 2: Mechanism: same-head counterfactual sensitivity $|p(x+\text{artifact}) - p(x)|$ versus the artifact’s overlap with the ROI in the five controlled sweeps (mean over seeds; band: range over seeds). Masking removes the out-of-ROI artifact ($r = 0$) but increasingly relies on an in-ROI one; erasure and balancing remove that reliance. Generated by `scripts/analysis/mechanism_figure.py`.

5.6. Chest radiography: a boundary case

With a synthetic tube swept into the lung fields of NIH pneumothorax radiographs (RAD-DINO), the encoder does not use an extra-pulmonary tube at all (reversed 0.913 versus clean 0.915 at no overlap), so masking has nothing to remove; on real devices (RANZCR-CLiP linked to NIH) masking helped at both locations, and real chest drains carry the shortcut mostly through correlates (the treated lung). We report chest radiography as the boundary of the law, not as a fifth confirmation.

6. Mechanism: masking retains an in-ROI artifact and removes its competitors

Three explanations were tested separately. *Occlusion*: with the ruler present but uncorrelated with the label (50/50), masking helps at every overlap (+0.036, +0.040, +0.039 at 0/50/100%); covering lesion pixels costs nothing measurable, so the harm requires the correlation. *Lost context*: dilating the mask at 50% overlap *increases* harm ($-0.097 \rightarrow -0.138 \rightarrow -0.154$ for margins 0/10/25 px) as the margin restores ruler pixels (retention $0.50 \rightarrow 0.73 \rightarrow 0.95$); at 100% overlap, where retention is already 1, harm is flat (-0.129 to -0.144) although the ruler’s share of visible pixels falls threefold. *Amplified reliance*: masking doubles the same-head counterfactual sensitivity to the ruler ($|\Delta p|$ $0.235 \rightarrow 0.491$ at 100%). Harm is largest for small lesions (-0.364 vs $+0.064$ for large lesions at 100%), where the ruler is a larger share of what survives the mask. Figure 2 shows the same retention mechanism in all five controlled sweeps: with the head held fixed, inserting the artifact changes the masked model’s prediction more the deeper the artifact lies in the ROI. At full overlap this sensitivity exceeds that of ERM for dermoscopic rulers (0.491 vs 0.235), ovarian calipers (0.218 vs 0.032) and capsule debris (0.302 vs 0.058) and matches it for thyroid calipers (0.393 vs 0.369). In chest radiography ERM ignores a tube outside the lungs ($|\Delta p| = 0.005$ at $r = 0$), which is why masking has nothing to remove there. Erasing a generic-overlay subspace after masking and group balancing both lower this sensitivity in every sweep (Supplement S8).

What the real-hair shortcut is made of. Removing hair pixels at test time only closes 44–47% of the ERM clean–reversed gap: most of what a frozen encoder exploits in a real hair trap is not the hair pixels but what accompanies hair (hair-bearing images differ in sex, age and body site). This bounds every pixel-level method, including oracle inpainting and U-MtE, and explains why image-level-label methods, which remove everything correlated with the label, recover more.

7. Why location matters: a linear-Gaussian account

Let a frozen encoder yield a disease block with class-separation signal-to-noise ratio S and an artifact block with SNR A per unit of artifact presence; training uses $P(A=1 \mid Y=1) = p_1$, $P(A=1 \mid Y=0) = p_0$ and the reversed test swaps them. For the LDA-optimal linear head the artifact’s effective training SNR is $\tilde{A} = (p_1 - p_0)^2 A / (1 + p_1(1 - p_1)A)$ and

$$\text{AUROC}_{\text{rev}}(S, \tilde{A}) = \Phi\left(\frac{S - \tilde{A}}{\sqrt{2(S + \tilde{A})}}\right), \quad (1)$$

increasing in S and decreasing in $\tilde{A}$ (verified by simulation to within 0.009 AUROC; Fig. 3). Masking maps $(S, \tilde{A}) \mapsto (S_m, \tilde{A}_m)$. Out of the ROI, $\tilde{A}_m = 0$; inside, $\tilde{A}_m \geq \tilde{A}$. Hence (i) in-ROI masking harms exactly when the mask removes disease-informative context ($S_m < S$) and helps only by denoising ($S_m > S$); (ii) for any S_m the benefit of masking is larger out of the ROI than inside it—the crossover is always positive; (iii) erasing a subspace that contains the artifact direction sets $\tilde{A} \rightarrow 0$ but scales S by $1 - \rho^2$, where ρ is its overlap with the disease direction, so erasure beats in-ROI masking unless the artifact resembles the pathology, which disease protection prevents; (iv) group balancing removes the artifact–label association and everything correlated with it, giving $\Phi(\sqrt{S/2})$ irrespective of location; (v) a shortcut carried by correlates that neither the mask nor pixel-level erasure touch can only be removed by label-based reweighting. Each consequence is borne out in Sections 5–9.

Does the account match real encoders? For any test composition $q_y = P(A=1 \mid Y=y)$ the same head gives $\text{AUROC} = \Phi(\Delta / \sqrt{\sigma_1^2 + \sigma_0^2})$ with $\Delta = S + (p_1 - p_0)(q_1 - q_0)A / (1 + vA)$, $\sigma_y^2 = S + w_a^2(1 + Aq_y(1 - q_y))$, $w_a = \sqrt{A}(p_1 - p_0) / (1 + vA)$ and $v = [p_1(1 - p_1) + p_0(1 - p_0)]/2$. In a pre-registered test we fitted (S, A) for every trained model to its clean and correlated AUROC *only* and compared the implied reversed-test AUROC with the held-out observation; no parameter is shared across models or tuned (Fig. 4). Two reference predictors use the same two inputs: “reversed = clean” (no shortcut harm) and a linear-symmetry heuristic, reversed = 2·clean – correlated. The endpoint on which the account is most informative is the held-out reversed-test AUROC itself: over 148 ERM and masking models (real traps, chest drains, controlled sweeps) its mean absolute error is 0.039 AUROC ($r = 0.92$), against 0.108 for the symmetry heuristic and 0.234 for “reversed = clean”. Along the controlled overlap sweeps it recovers the sign of the masking gain at

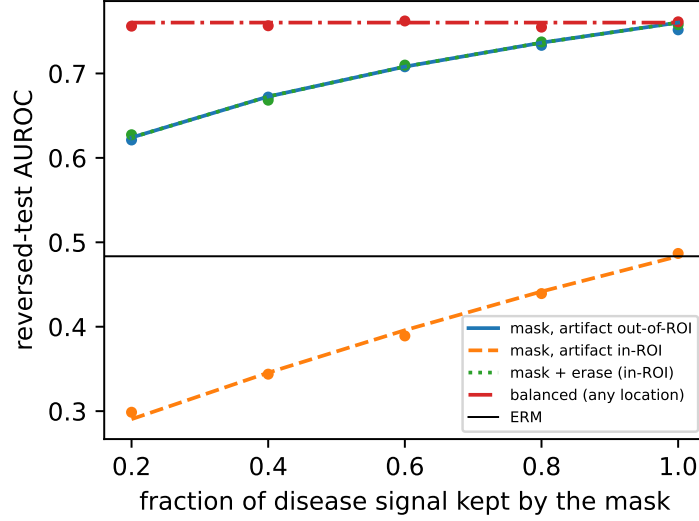


Figure 3: Reversed-test AUROC predicted by the linear-Gaussian model (lines) and simulated with logistic heads (points) as masking removes a growing share of disease signal ($S = 1$, $A = 2$, 90/10 training). Generated by `scripts/analysis/theory_sim.py`.

42 of 45 overlap points (symmetry heuristic: 36/45). For the location crossover the gain over the heuristic is small: both follow the observed crossovers closely across 14 cohort–backbone pairs (account: $r = 0.997$, mean absolute error 0.014, 13/14 signs; heuristic: $r = 0.989$, error 0.036, 13/14), because a crossover is a difference of masking gains in which most of the symmetry already cancels. We therefore do not use the crossover correlation as evidence for the model. The account also matches end-to-end fine-tuned networks (4/4 crossover signs), which violate its assumptions, and, fitted to DermLIP’s clean and correlated AUROC, it anticipates that encoder’s absence of a hair location gap (-0.003 implied, -0.002 observed; Sec. 5.5). Because our pre-registered criterion required every crossover sign to agree (one chest-radiograph cell with an observed crossover of -0.002 disagrees), we call the model a qualitative account. Its one clear failure is the chest-drain cohort (implied 0.70–0.76, observed 0.19–0.24): the clean and correlated environments do not expose a shortcut carried by correlates of the drain, regime (v).

8. Consequences on unaltered test data

The traps construct an extreme reversal. Without any resampling, in-ROI artifacts are still associated with the label in every cohort (thyroid calipers: 37.6% of benign vs 24.1% of malignant nodules; in-lesion hair: 31% of melanomas vs 20% of benign lesions; capsule debris on the lesion: 23% of polyp-like lesions vs 10% of erosions). We trained on natural training data and tested on untouched data: the official TN3K/TNCD test split, which is patient-disjoint from the training split [12], each ISIC source held out in turn (a real

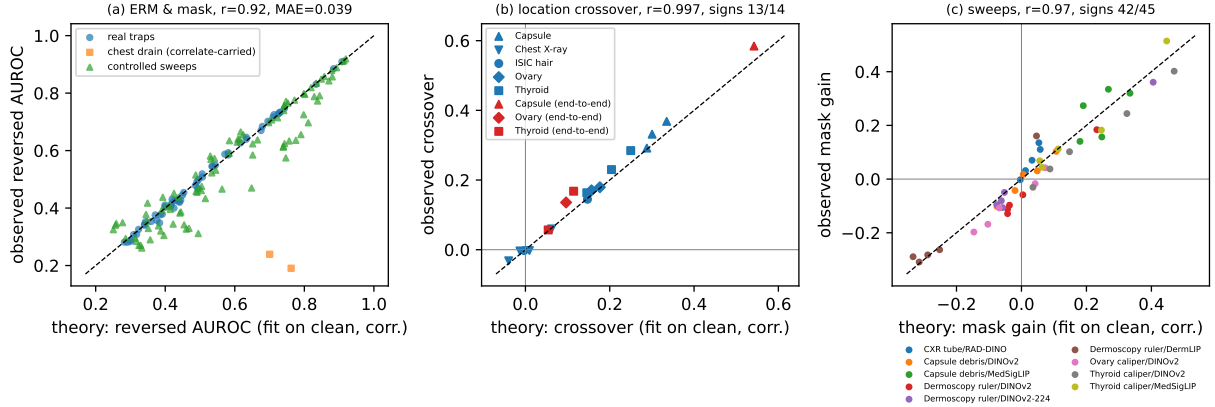


Figure 4: Pre-registered test of the linear-Gaussian account on real encoders. (S, A) are fitted to each model’s clean and correlated AUROC; the reversed-test results are held out. (a) Implied vs observed reversed-test AUROC (ERM and masking). (b) Location crossover per cohort and backbone; red: end-to-end fine-tuned networks, outside the model’s assumptions. (c) Mask – ERM gain along the controlled overlap sweeps. Dashed: identity. Generated by `scripts/analysis/theory_predict.py`.

change of hospital and device), and held-out capsule frame blocks.

Discrimination. On the shortcut-conflicting pairs, masking *lowers* AUROC for thyroid (-0.102 [$-0.139, -0.066$]) and for two of the three held-out hospitals (BCN -0.016 [$-0.027, -0.005$], MSK -0.056 [$-0.084, -0.029$]); it helps for HAM ($+0.053$ [$+0.037, +0.068$]) and does not differ significantly from ERM on capsule. Over all test pairs masking changes thyroid AUROC by only -0.005 [$-0.027, +0.018$]: the harm is confined to the cases in which the in-ROI artifact points to the wrong diagnosis, which is exactly where a shortcut is dangerous.

Operating points (pre-registered). Because a clinical decision uses a threshold, we fixed thresholds on each seed’s validation split at five operating points and compared test sensitivity and specificity (Table 3; the bootstrap re-estimates the thresholds in every replicate). On the official thyroid split masking lowers sensitivity at all five operating points, each with a CI excluding zero, for example from 0.699 to 0.424 at the maximal-balanced-accuracy threshold (-0.275 [$-0.399, -0.182$]) and from 0.557 to 0.355 at a validation specificity of 0.80 (-0.202 [$-0.265, -0.134$]). Specificity rises correspondingly, so part of the change is a shift of the score distribution; but the loss is concentrated where the shortcut conflicts with the label: among malignant nodules that carry an in-ROI caliper, at a validation specificity of 0.80 sensitivity falls by -0.372 [$-0.436, -0.291$] ($0.549 \rightarrow 0.177$). The pre-registered decision rule (loss at the maximal-balanced-accuracy threshold and at ≥ 3 of the four other operating points) is met. On capsule endoscopy, where masking improves AUROC overall, it lowers the sensitivity for erosions carrying debris on the lesion at the two high-sensitivity operating points (Supplement S9).

Table 3: Masking versus ERM on the unaltered official thyroid test split at five operating points fixed on validation data (mean over five seeds; Δ = mask – ERM with 95% hierarchical bootstrap CI, thresholds re-estimated in every replicate). *Conflicting*: malignant nodules with an in-ROI caliper (the caliper marks benign nodules in this dataset) and benign nodules without one.

Threshold on validation	Sensitivity ERM $\rightarrow$ mask	Δ sensitivity	Δ sensitivity, conflicting	Δ specificity
max. balanced accuracy	0.699 $\rightarrow$ 0.424	−0.275 [−0.399, −0.182]	−0.479 [−0.583, −0.362]	+0.202 [+0.140, +0.300]
specificity $\geq$ 0.80	0.557 $\rightarrow$ 0.355	−0.202 [−0.265, −0.134]	−0.372 [−0.436, −0.291]	+0.117 [+0.082, +0.151]
specificity $\geq$ 0.90	0.371 $\rightarrow$ 0.219	−0.152 [−0.225, −0.071]	−0.236 [−0.345, −0.132]	+0.075 [+0.043, +0.109]
sensitivity $\geq$ 0.80	0.753 $\rightarrow$ 0.380	−0.373 [−0.431, −0.310]	−0.546 [−0.614, −0.480]	+0.278 [+0.229, +0.329]
sensitivity $\geq$ 0.90	0.869 $\rightarrow$ 0.588	−0.281 [−0.353, −0.221]	−0.462 [−0.560, −0.369]	+0.339 [+0.269, +0.396]

Calibration. For the held-out BCN hospital masking worsens both Brier score and expected calibration error (Table 4); for HAM, MSK and capsule it improves every metric, in line with the AUROC results.

9. What carries the shortcut decides the fix

The location law says what masking cannot do; it does not by itself say what to do instead. We evaluated the mitigation families that a practitioner can apply to a frozen encoder, grouped by the supervision they need (Table 5; full results, methods and ablations in Supplement S8). Three regimes emerge, and the linear-Gaussian account (Sec. 7) explains each.

(1) *The artifact is a distinct overlay that carries the shortcut in its pixels.* (sonographer calipers; synthetic rulers, debris and tubes). Masking removes an out-of-ROI copy and leaves an in-ROI copy. Without any artifact annotation, erasing from the frozen features the subspace along which *generic* synthetic overlays move the representation of masked images (mask-then-erase, U-MtE; Supplement S8) removes most of what masking leaves: thyroid +0.222 [+0.197, +0.251] over masking with DINOv2, +0.204 [+0.174, +0.237] with MedSigLIP and +0.128 [+0.112, +0.145] with a ConvNeXt, and erasure beats augmentation with the same overlays (thyroid +0.207 [+0.182, +0.235]). With image-level artifact labels, group balancing on top of it is the most robust option (Table 5).

(2) *The artifact resembles the pathology.* (capsule debris versus erosion fibrin). Any subspace estimated from artifact appearance also contains disease evidence (ρ large in Sec. 7); unprotected erasure is worse than masking (−0.115), protecting the label directions removes the failure but not the shortcut (−0.010 [−0.020, −0.001]). With image-level artifact labels masking plus last-layer retraining is the most robust option (0.88).

(3) *The shortcut is carried by correlates of the artifact.* (dermoscopic hair co-varies with body site, age and sex; a chest drain comes with a treated lung). Removing every hair patch with an oracle mask gains only +0.015; no pixel- or insertion-based method closes the gap, and label-based reweighting (DFR, group balancing) is required. The oracle-removal ceiling on a handful of annotated images diagnoses this regime.

Table 4: Clinical secondary metrics on the unaltered test sets (Sec. 8): mean (SD) over training seeds. Sensitivity and specificity at the operating point fixed on clean validation data; ECE with 10 equal-width bins; AUPRC baseline = prevalence.

Test set	Arm	AUROC	AUPRC	Brier	ECE	Sens	Spec
Thyroid (official split) ($\pi = 0.38$)	ERM	0.736 (0.009)	0.606 (0.012)	0.207 (0.005)	0.082 (0.008)	0.734 (0.047)	0.619 (0.061)
	mask	0.731 (0.006)	0.632 (0.007)	0.214 (0.006)	0.120 (0.026)	0.347 (0.095)	0.888 (0.041)
	U-MtE	0.742 (0.003)	0.638 (0.007)	0.203 (0.001)	0.085 (0.007)	0.425 (0.088)	0.854 (0.050)
	balanced	0.716 (0.007)	0.584 (0.012)	0.219 (0.004)	0.110 (0.007)	0.714 (0.052)	0.599 (0.061)
	U-MtE _{bal}	0.731 (0.004)	0.630 (0.007)	0.207 (0.001)	0.084 (0.006)	0.399 (0.069)	0.866 (0.034)
ISIC, BCN held out ($\pi = 0.23$)	ERM	0.786 (0.003)	0.586 (0.003)	0.239 (0.006)	0.279 (0.011)	0.842 (0.038)	0.519 (0.075)
	mask	0.769 (0.004)	0.571 (0.009)	0.286 (0.007)	0.332 (0.009)	0.860 (0.013)	0.439 (0.023)
	U-MtE	0.773 (0.004)	0.574 (0.008)	0.280 (0.009)	0.328 (0.012)	0.858 (0.039)	0.448 (0.089)
	balanced	0.781 (0.005)	0.585 (0.009)	0.228 (0.006)	0.257 (0.011)	0.811 (0.046)	0.554 (0.087)
	U-MtE _{bal}	0.774 (0.004)	0.568 (0.008)	0.279 (0.011)	0.325 (0.016)	0.860 (0.027)	0.447 (0.051)
ISIC, MSK held out ($\pi = 0.19$)	ERM	0.770 (0.008)	0.477 (0.012)	0.172 (0.007)	0.161 (0.019)	0.608 (0.049)	0.786 (0.039)
	mask	0.790 (0.004)	0.512 (0.009)	0.146 (0.003)	0.115 (0.009)	0.593 (0.052)	0.824 (0.037)
	U-MtE	0.783 (0.004)	0.500 (0.008)	0.150 (0.003)	0.126 (0.009)	0.566 (0.073)	0.831 (0.049)
	balanced	0.766 (0.008)	0.462 (0.014)	0.178 (0.008)	0.170 (0.020)	0.626 (0.047)	0.755 (0.046)
	U-MtE _{bal}	0.774 (0.006)	0.482 (0.009)	0.154 (0.004)	0.131 (0.010)	0.594 (0.061)	0.798 (0.051)
ISIC, HAM held out ($\pi = 0.11$)	ERM	0.750 (0.005)	0.353 (0.005)	0.162 (0.006)	0.234 (0.011)	0.535 (0.092)	0.813 (0.068)
	mask	0.789 (0.002)	0.404 (0.006)	0.143 (0.005)	0.217 (0.011)	0.559 (0.112)	0.842 (0.063)
	U-MtE	0.799 (0.008)	0.406 (0.009)	0.145 (0.007)	0.226 (0.016)	0.573 (0.067)	0.846 (0.041)
	balanced	0.747 (0.017)	0.346 (0.019)	0.156 (0.014)	0.228 (0.015)	0.539 (0.078)	0.803 (0.076)
	U-MtE _{bal}	0.802 (0.005)	0.401 (0.009)	0.139 (0.006)	0.215 (0.021)	0.606 (0.054)	0.825 (0.037)
Capsule (held-out frames) ($\pi = 0.56$)	ERM	0.889 (0.019)	0.897 (0.046)	0.130 (0.016)	0.042 (0.011)	0.770 (0.062)	0.844 (0.055)
	mask	0.967 (0.011)	0.971 (0.014)	0.069 (0.013)	0.029 (0.011)	0.859 (0.032)	0.933 (0.025)
	U-MtE	0.962 (0.009)	0.967 (0.013)	0.081 (0.017)	0.065 (0.045)	0.871 (0.042)	0.897 (0.043)
	balanced	0.881 (0.024)	0.889 (0.053)	0.136 (0.017)	0.051 (0.009)	0.758 (0.087)	0.838 (0.059)
	U-MtE _{bal}	0.959 (0.008)	0.964 (0.012)	0.079 (0.007)	0.044 (0.012)	0.866 (0.048)	0.897 (0.052)

Remedies do not transfer to unaltered test data. The regimes above are defined on the traps, whose reversed test sets are extreme by construction. On the unaltered test distributions (Sec. 8) no remedy is uniformly better than the unmasked model: annotation-free U-MtE is flat or negative (capsule -0.005 $[-0.010, +0.000]$, held-out BCN hospital $+0.004$ $[+0.002, +0.006]$ over masking), its protected variant costs -0.022 $[-0.026, -0.018]$ on capsule, and group balancing can hurt (-0.086 $[-0.106, -0.067]$ on capsule). We therefore offer the regimes as a decision guide for *auditing* a model, not as a claim that any remedy improves deployment performance: (i) measure where artifacts sit relative to the ROI; out-of-ROI artifacts are what masking removes; (ii) for in-ROI artifacts estimate which regime applies (oracle-removal ceiling, resemblance to the pathology); (iii) evaluate any mitigation on shortcut-conflicting cases *and* on the unaltered distribution, reporting clean, correlated and reversed performance together, because a reversed-environment gain can be manufactured by removing diagnosis signal (Supplement S8).

Table 5: In-ROI artifacts (Trap A): robustness = min(reversed, correlated) AUROC, which penalises both shortcut use and shortcut flipping. Bold: best per column. Full reversed/correlated/clean values for all cohorts and backbones: Table ??.

Method	Needs	Hair DINOv2	Thyroid DINOv2	Thyroid MedSigLIP	Ovary DINOv2	Capsule DINOv2	Drain RAD-DINO
ROI masking	ROI	0.45	0.40	0.35	0.55	0.68	0.24
Mask + overlay augmentation	ROI	0.46	0.42	0.38	0.53	0.70	0.24
U-MtE (ours)	ROI	0.51	0.62	0.55	0.56	0.57	0.27
U-MtE, protected (ours)	ROI	0.49	0.62	0.61	0.59	0.67	0.27
Group-balanced	A	0.72	0.68	0.70	0.71	0.78	0.54
DFR	A	0.74	0.66	0.72	0.69	0.80	0.68
Mask + DFR	ROI + A	0.69	0.74	0.76	0.75	0.88	0.65
U-MtE + balanced (ours)	ROI + A	0.70	0.79	0.79	0.74	0.86	0.46

10. Robustness and reproducibility

10.1. Robustness to artifact-label definitions

Artifact masks for calipers come from a detector (visual audit: 95% of thyroid detections lie on real calipers; 92–97% of caliper-free images contain no visible caliper). Re-running the thyroid and ovary analyses with large-marker-only (≥ 50 px) and strict-location ($r \geq 0.7$ / $r < 0.05$) definitions leaves the crossover (thyroid +0.215 to +0.262; ovary +0.170 to +0.195) and the U-MtE_{protect} gain over masking (thyroid +0.217 to +0.282; ovary +0.043 to +0.067) significant in every variant; both grow under stricter definitions, as expected if detector errors dilute rather than create the effects.

10.2. Stricter leakage grouping

TN3K/TNCD, MMOTU and SEE-AI publish no patient or video identifiers, so the main analyses group near-duplicates by perceptual hash (and, for capsule, blocks of 100 consecutive frames). As a pre-registered sensitivity analysis we united these groups with connected components of mean-centred DINOv2 embeddings at the most aggressive cosine threshold that keeps every group below 5% of the cohort (thyroid: 3,455 $\rightarrow$ 3,081 groups, $\tau = 0.80$; ovary: 1,059 $\rightarrow$ 969, $\tau = 0.80$; capsule: 137 $\rightarrow$ 135, $\tau = 0.93$) and re-ran all primary contrasts. Eleven of twelve keep their sign and significance: the location crossover is +0.235 [+0.211, +0.259] (thyroid), +0.157 [+0.080, +0.235] (ovary) and +0.394 [+0.359, +0.430] (capsule), and protected U-MtE still beats masking on thyroid (+0.251 [+0.219, +0.279]) and ovary (+0.047 [+0.020, +0.072]). The exception is the small ovary advantage of erasure over augmentation, which loses significance (+0.020 [−0.011, +0.053]). Embedding grouping removes visually near-identical images of one patient or session; it cannot exclude differently looking images of the same patient.

10.3. Rebuild on new hardware

All data were downloaded again and every derived object—caches, perceptual-hash groups, the retrained lesion U-Net (held-out Dice 0.886; HAM10000 agreement 0.895), the caliper scan and the debris probe (IoU 0.625)—was rebuilt on a different GPU generation (pre-registered). Trap membership changes slightly where it depends on a retrained model (hair traps: Trap A 2,655/942 artifact-bearing benign/melanoma images instead of 2,738/989); the thyroid and ovary cells are identical. Every primary crossover reproduces within the pre-registered tolerance: dermoscopic hair +0.147 [+0.119, +0.177] (archived +0.145); thyroid +0.239 [+0.212, +0.264] (archived +0.230); ovary +0.170 [+0.116, +0.223] (archived +0.170); capsule +0.377 [+0.339, +0.412] (archived +0.368).

11. Discussion and limitations

What the evidence supports. Masking is not a neutral preprocessing step: its benefit depends on where the artifact sits. The strongest evidence is the real-artifact transplant, in which the same real calipers, hair or debris on the same images are removed by masking outside the ROI and retained—with amplified reliance—inside it; the real traps, covariate-matched traps, controlled sweeps, fine-tuned networks and foundation-model zero-shot scores all point the same way. The claim is about the *gap*. Whether in-ROI masking helps a little or harms depends on whether the mask also removes disease context, which the linear-Gaussian account makes explicit and which differs between modalities (harm in dermoscopy, small gains in ultrasound and capsule endoscopy).

Why this matters clinically. On the patient-disjoint official thyroid test split the in-ROI caliper is associated with benign nodules; masking keeps it and removes competing evidence, and the malignant nodules that carry a caliper are then missed more often at every operating point we examined. An ROI-masked model can therefore look unchanged by overall AUROC while failing on exactly the cases in which the shortcut points the wrong way. Audits of masked models should stratify performance by artifact presence and location.

Reading reversed-environment gains. A reversed-test gain can be manufactured by removing diagnosis-correlated signal together with the artifact: erasure fitted on unpaired image-level labels and post-hoc prevalence recalibration invert the correlated/reversed ordering. We report reversed, correlated and clean AUROC for every method and summarise robustness by the minimum of correlated and reversed AUROC.

Limitations. (i) *Artifact labels.* Calipers are segmented by a rule-based detector and capsule debris by a patch-token probe (IoU 0.62 against expert masks); hair masks are published semi-automatic masks. Our visual audits were carried out by the analysis team, not by clinicians; an expert audit with two independent raters and inter-rater agreement is prepared (blinded kit and scoring script in the repository) but not yet

done. Label errors dilute the constructed correlation and should bias effects toward zero, and stricter label definitions strengthen rather than weaken the results (Sec. 10); the transplant experiment does not depend on the artifact-free group being perfectly clean because the transplanted instance is identical at both locations.

(ii) *Patient identifiers.* TN3K/TNCD, MMOTU and SEE-AI publish no patient or video identifiers for the pooled images, so the trap splits group near-duplicates by perceptual hash and frame blocks (a stricter embedding grouping leaves 11 of 12 primary estimates unchanged); the official TN3K/TNCD test split used for the unaltered-data analysis is patient-disjoint. The one open cohort we found with patient identifiers, pathology labels and real calipers (ThyUS2Path; 842 patients) has no nodule masks and uses a scanner interface on which our frozen caliper detector fires on interface elements; external validation on it requires a transferred nodule segmentation and a re-validated detector and is left for future work.

(iii) *Constructed shifts.* Reversed environments are constructed by subsampling; real deployment shifts are less extreme, which is why we report the unaltered test distributions separately.

(iv) *Absolute performance.* Frozen encoders with linear heads are below the best published fine-tuned models on the one cohort with a directly comparable benchmark (official TNCD split: 0.736 AUROC versus 0.773–0.784 for fine-tuned CNNs with cross-entropy and up to 0.799 with curriculum learning, 12). The law replicates with end-to-end fine-tuned networks in every cohort (Sec. 5.5), and the other cohorts’ binary tasks have no published benchmark on the same split (Supplement S10).

(v) *Scope.* Chest radiography is a boundary case: the encoder ignored extra-pulmonary tubes and real drains carry the shortcut through correlates. The linear-Gaussian account is fitted per model and cannot anticipate a shortcut carried by correlates that the clean and correlated environments do not expose.

(vi) *Pre-registration.* Every registration is committed before its outcome results (registry with commit hashes and times in Supplement S1); commit times are set by the committing machine, so only the registrations of the second review round carry an external (GitHub push) time stamp, and the 16-test primary family was assembled after the individual results.

12. Conclusion

ROI masking is not a neutral preprocessing step. Its benefit depends on where the artifact sits: it removes shortcuts that live outside the ROI and leaves—or amplifies—those that live inside it. This held in four diseases, three modalities, three frozen backbones, two fine-tuned architectures and on unaltered data, and it follows from a two-parameter model of signal and artifact strength. For in-ROI artifacts, erasing a subspace learned from generic synthetic overlays removes the shortcut when the artifact is a distinct overlay, needs no annotation of the artifact, and should be disease-protected; when the shortcut travels with correlates of the artifact, reweighting with image-level artifact labels is required. We recommend that shortcut audits report artifact location relative to the ROI and evaluate mitigations on shortcut-conflicting cases, alongside clean and correlated performance.

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